{EPITECH}

HACKATHON MOBILITY_

< FROM OPEN DATA TO LOCATION DECISIONS />



HACKATHON MOBILITY

Problem statement

Mapping access to shared mobility: who actually has a transport option near home, and where should more be added?

Description

Buses, trams, bike and scooter sharing, car sharing: mobility options keep multiplying, but they are very unevenly spread across the territory.

Since the French mobility orientation law (LOM), a large share of these services publish their data as open data. Public transport networks use the **GTFS** format, shared services (bikes, scooters, car sharing) use the **GBFS** format. Population data is public as well.

Citiz, partner of this hackathon, is the leading cooperative car-sharing network in France: nearly 2,800 shared cars, spread across about 280 cities and used by nearly 130,000 drivers. Like any mobility operator, Citiz has to choose where to open its next stations: where they will reach the most residents, complement the existing offer and find their users. The overall goal of the hackathon is to turn this open data into a **decision-support tool for choosing where to set up**.

Everything is therefore available to answer a simple question: who actually has access to a mobility option near home, and where should more be added?

The challenge

Using open data (GBFS, transport.data.gouv.fr or any other public source), you must meet the following three objectives, on French territory, overseas departments and regions (DROM) included. You may narrow the analysis to a metropolitan area or a region, as long as you justify that choice. Citiz may not yet be present in your region: in that case, the third objective is to show where and how Citiz could set up there.

1. **Map access to mobility in France.**
   Produce a map where each area is colored, from coldest to hottest, according to the number of transport options reachable per inhabitant within a 5-minute walk.
2. **Show who is concerned.**
   Display on the map the different types of stations and stops, along with the number of people each one serves.
3. **Propose where to set up and invest.**
   Identify the locations where Citiz should set up new car-sharing stations, and more broadly where to add a bus stop or a bike-sharing station, to reach the most people at the lowest cost.

The third objective is the core of the project. The first two build the diagnosis, the third turns it into a decision.

Prediction and optimization tool

For the third objective, you must design a **prediction and optimization tool** that:

- ✓ estimates, for a candidate location, the value of a new station or stop (newly served population, potential usage, etc.) using a **predictive model**;
- ✓ includes a **cost** estimate for each type of facility;
- ✓ proposes a **prioritized list of locations** under a budget constraint, taking the existing offer into account.

To build this tool, we suggest the **Microsoft tools** provided through the partnership, **Microsoft Fabric** in particular, whether for data processing, machine learning or visualization. How to access them will be communicated at the start of the hackathon. You remain free to use other tools, as long as you justify that choice during the defense.

Data to use

No single source is enough. The transport offer says nothing about the population served, today's population says nothing about tomorrow's needs, and a map without the pedestrian network overestimates actual accessibility. Combining several sources, of different nature and scale, is what turns a simple inventory into a credible location recommendation.

- ✓ Public transport stops and lines (GTFS format).
- ✓ Shared stations and vehicles: bikes, scooters, car sharing (GBFS format).
- ✓ Resident population at a fine scale (200 m grid squares, IRIS or municipalities).
- ✓ Demographic projections, to anticipate changes in population and age structure.
- ✓ Pedestrian or road network, if you compute actual walking distances.
- ✓ Any other relevant data: income, household car ownership, amenities, jobs, etc.

The sources

- ✓ National access point for transport data:
  `https://transport.data.gouv.fr/datasets`
  - Public transit (GTFS)
    `https://transport.data.gouv.fr/datasets?type=public-transit`
  - Shared vehicles (GBFS)
    `https://transport.data.gouv.fr/datasets?type=vehicles-sharing`
  - Citiz car sharing, partner of the hackathon (GBFS feeds per regional network)
    `https://transport.data.gouv.fr/datasets?type=vehicles-sharing&q=citiz`

  - Transport stops in France (national aggregate of GTFS stops, not deduplicated)
    https://www.data.gouv.fr/datasets/arrets-de-transport-en-france
- ✓ Worldwide catalog of GBFS feeds (MobilityData):
  https://github.com/MobilityData/gbfs/blob/master/systems.csv
- ✓ Population per 200 m grid square (INSEE, Filosofi):
  https://www.insee.fr/fr/statistiques/8735162
- ✓ Population projections 2018-2070 by department and region (INSEE, Omphale):
  https://www.insee.fr/fr/statistiques/6652134
- ✓ Administrative and IRIS boundaries (IGN):
  https://geoservices.ign.fr/adminexpress
- ✓ Pedestrian and road network (OpenStreetMap):
  https://download.geofabrik.de/europe/france.html

Any other relevant public source is welcome, as long as it is cited.

Format references

- ✓ GTFS: https://gtfs.org/
- ✓ GBFS: https://gbfs.org/

Partners

This hackathon is organized in partnership with:

- ✓ **Citiz**, a cooperative car-sharing network, whose stations are part of the data to analyze;
- ✓ **Microsoft**, which provides its data processing and artificial intelligence tools, including Microsoft Fabric.

Target audience

- ✓ Local authorities and mobility organizing authorities.
- ✓ Public transport and shared-mobility operators.
- ✓ Urban planners, researchers and data scientists.

Deliverables

- ✓ The code of your project, on a Git repository.
- ✓ The slides of your presentation, on the same repository.
- ✓ The list of data sources used, on the same repository.

Teams selected for the national finals also provide a functional prototype and technical documentation allowing the work to be continued.

Evaluation

You have two and a half days to develop your solution, but above all to **convince**. The evaluation is based on a defense, with:

- ✓ a **functional demonstration**, live, on real data;
- ✓ an argument that is **convincing for a non-technical person**: a concrete problem, a clear recommendation, a quantified impact, and if possible a look at what the recommendation offers an operator such as Citiz;
- ✓ an argument that is **convincing for a data scientist**: justified methodological choices, a rigorously evaluated model, acknowledged limits.

Every team member must be able to explain the overall approach of the project and to answer in depth on the part they contributed to.

How you handle the ethical and societal issues of your recommendation is also part of the evaluation, for example:

- ✓ the estimated impact on the city's carbon footprint (shift from private car trips to shared mobility);
- ✓ equity between dense and sparsely populated areas, and the risk of serving only areas that are already well served;
- ✓ the risk of stigmatizing a territory by labeling it as ”poorly served”;
- ✓ privacy protection in very sparsely populated areas.

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